

Every Algebraic Number is a Critical Value of an Integer Polynomial

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Abstract

For every nonconstant $f \in \mathbb{Z}[X]$ we construct $g \in \mathbb{Z}[X]$ and an irreducible $H \in \mathbb{Z}[X]$ with $H \mid g'$ and $H^2 \mid f \circ g$. Equivalently, every algebraic number is a critical value of an integer polynomial, so the set of critical values of $\mathbb{Z}[X]$ is exactly $\overline{\mathbb{Q}}$. The construction is explicit, and the proof is root-free: no algebraic closure, no minimal polynomials and no splitting fields appear in it, only two exact identities in $\mathbb{Z}[X]$ and an induction on the prime factors of an integer. Algebraic numbers enter only in the statement of the corollary. The argument has been formalized in Lean 4 and the construction implemented as a program.

1 Introduction

A critical value of $g \in \mathbb{Z}[X]$ is a number α with $g(\beta) = \alpha$ and $g'(\beta) = 0$ for some β . Since $g' \neq 0$ in characteristic zero, its roots are algebraic and g carries algebraic numbers to algebraic numbers, so every critical value of every nonconstant $g \in \mathbb{Z}[X]$ is algebraic. This note proves the converse, and does so with an explicit construction.

Theorem 1. *Let $f \in \mathbb{Z}[X]$ be nonconstant. Then there exist $g, H \in \mathbb{Z}[X]$ with H irreducible in $\mathbb{Z}[X]$ and $1 \leq \deg H$, such that*

$$H \mid g' \quad \text{and} \quad H^2 \mid f \circ g \quad \text{in } \mathbb{Z}[X].$$

The g produced satisfies $\deg g \geq 2$.

Two corollaries follow at once. The first is the divisibility statement in the form one usually wants it.

Corollary 1. *With g as in Theorem 1, write $g' = c \prod_i g_i$ with each g_i irreducible in $\mathbb{Z}[X]$ and $c \in \mathbb{Z}$. Then $g_i^2 \mid f \circ g$ for at least one i .*

Proof. H is irreducible, hence prime in the unique factorization domain $\mathbb{Z}[X]$, and $\deg H \geq 1$, so $H \nmid c$ (as a constant polynomial). From $H \mid c \prod_i g_i$ we get $H \mid g_i$ for some i ; two irreducibles with one dividing the other are associates, and associates have associate squares, so $g_i^2 \mid H^2 \mid f \circ g$. $\square$

The second is the statement in the title.

Corollary 2. *For every $\alpha \in \overline{\mathbb{Q}}$ there are $g \in \mathbb{Z}[X]$ and $\beta \in \overline{\mathbb{Q}}$ with $g(\beta) = \alpha$ and $g'(\beta) = 0$. Consequently the set of critical values of nonconstant integer polynomials is exactly $\overline{\mathbb{Q}}$.*

The point of the proof below is what it does *not* use. One expects a question about algebraic numbers to be answered by working in a number field: choose α , take its minimal polynomial, look for a critical point among the conjugates. That route runs into the unit group and the class group of $\mathbb{Q}(\alpha)$ almost immediately, and into the question of whether $\mathcal{O}_K$ is monogenic. None of that is needed. The construction is a single pair of polynomial identities, valid in $\mathbb{Z}[X]$ with no reference to any root of anything, followed by an elementary descent that removes an unwanted integer factor. Roots appear exactly once, in §7, to convert the divisibility statement into the statement about critical values.

There is one structural warning, which §8 makes precise: H cannot be required to be monic. The monic form of Theorem 1 is false, and the obstruction is not a matter of a harder search.

1.1 Conventions

Throughout, $p' = \text{derivative } p$, and for a fixed nonzero $M \in \mathbb{Z}$ we write

$$\tilde{p} := p(MX) = p \circ (MX),$$

the *dilation* of p . Dilation is a ring homomorphism $\mathbb{Z}[X] \rightarrow \mathbb{Z}[X]$, it is injective for $M \neq 0$, it preserves degree, and it interacts with the derivative by the chain rule

$$(\tilde{p})' = M \tilde{p}'. \quad (1)$$

These four facts are used on essentially every line of §3 and are worth isolating before anything else; the factor M in (1) is the reason M has to appear in the statement of Lemma 2.

2 The construction

Fix a nonconstant $f \in \mathbb{Z}[X]$ and suppose $X \nmid f$; the excluded case is settled separately in §6. All of the following data is explicit and computable.

<i>Symbol</i>	<i>Definition</i>	<i>Notes</i>
h	a primitive irreducible factor of f with $h(0) \neq 0$	in $\mathbb{Z}[X]$
f_1	the cofactor, $f = hf_1$	in $\mathbb{Z}[X]$
n	$\deg h \geq 1$	
h_0	$h(0) \neq 0$	in $\mathbb{Z}$
u, v, ρ	any Bézout data, $uh + vh' = \rho$, $\rho \neq 0$	$u, v \in \mathbb{Z}[X]$, $\rho \in \mathbb{Z}$
M	ρh_0 , so $M \neq 0$	the dilation factor
H	$h(MX)/h_0$	$H(0) = 1$, $\deg H = n$
W	$(v(0)H - v(MX))/\rho$	$W(0) = 0$
g	$MX + h(MX)W = MX + h_0HW$	

Existence of h . f is nonconstant, so it has an irreducible factor. Irreducible factors of positive degree in $\mathbb{Z}[X]$ are primitive, and $h(0) \neq 0$ because $h(0) = 0$ would give $X \mid h \mid f$, excluded.

Existence of the Bézout data. h is irreducible and primitive in $\mathbb{Z}[X]$, hence irreducible in $\mathbb{Q}[X]$ by Gauss's lemma. In characteristic zero $h' \neq 0$ and $\deg h' < \deg h$, so $h \nmid h'$ and therefore $\gcd(h, h') = 1$ in the principal ideal domain $\mathbb{Q}[X]$. Bézout there and clearing denominators produces $u, v \in \mathbb{Z}[X]$ and $\rho \in \mathbb{Z} \setminus \{0\}$.

Any nonzero ρ in the elimination ideal $(h, h') \cap \mathbb{Z}$ works. The resultant $\text{Res}(h, h')$ is one admissible choice and is never needed as such; smaller choices give a smaller g , and the generator of the elimination ideal — the *reduced* resultant — is the best available. Example 2 below uses $\rho = 2$ where the resultant is 4.

2.1 Integrality

The two divisions in the table are exact, which is the whole reason M was defined as the product ρh_0 rather than as either factor alone.

Lemma 1 (Integrality). $H \in \mathbb{Z}[X]$ with $H(0) = 1$ and $\deg H = n$; and $W \in \mathbb{Z}[X]$ with $W(0) = 0$.

Proof. Write $h = \sum_j h_j X^j$ and $v = \sum_j v_j X^j$. The coefficients of H are $H_0 = h_0/h_0 = 1$ and, for $j \geq 1$,

$$H_j = \frac{h_j M^j}{h_0} = \frac{h_j \rho^j h_0^j}{h_0} = h_j \rho^j h_0^{j-1} \in \mathbb{Z},$$

using $h_0 \mid M$. Since $\deg h = n$ and $M \neq 0$, $\deg H = n$.

For W , the coefficient of X^j in $\rho W = v(0)H - \tilde{v}$ is $v(0)H_j - v_j M^j$. At $j = 0$ this is $v_0 \cdot 1 - v_0 = 0$, which gives both integrality there and $W(0) = 0$. For $j \geq 1$ the display above exhibits $\rho^j \mid H_j$, and $\rho \mid M^j$, so ρ divides the whole coefficient. $\square$

Remark 1. For the proof of Theorem 1 it is cleaner to treat H and W as *data* subject to the defining identities

$$h_0 H = \tilde{h}, \quad \rho W = v(0)H - \tilde{v}, \quad H(0) = 1, \quad W(0) = 0, \quad (2)$$

rather than as quotients. Nothing after this point uses anything else about them, and no division in $\mathbb{Z}[X]$ ever has to be justified again. Lemma 1 is then exactly the statement that such data exists.

3 The two identities

Both statements below are exact identities in $\mathbb{Z}[X]$. There are no congruences, no roots and no primality anywhere in this section; in particular *irreducibility of h is never used here*. It is needed only to produce the Bézout data in §2 and again in §5.2.

Dilating the Bézout relation $uh + vh' = \rho$ by the ring homomorphism $p \mapsto \tilde{p}$ gives the form in which it will be used,

$$\tilde{u} \tilde{h} + \tilde{v} \tilde{h}' = \rho, \quad \text{equivalently} \quad \tilde{v} \tilde{h}' = \rho - h_0 \tilde{u} H, \quad (3)$$

the second form by $\tilde{h} = h_0 H$.

Lemma 2 (Derivative). $\rho g' = HA$, where $A := Mh_0 \tilde{u} + Mv(0) \tilde{h}' + \rho h_0 W' \in \mathbb{Z}[X]$.

Proof. From $g = MX + \tilde{h}W$ and the chain rule (1),

$$g' = M + M \tilde{h}' W + \tilde{h} W'.$$

Multiply by ρ and substitute $\rho W = v(0)H - \tilde{v}$ and $\tilde{h} = h_0 H$:

$$\rho g' = \rho M + M \tilde{h}' (v(0)H - \tilde{v}) + h_0 H \rho W' = \rho M - M \tilde{h}' \tilde{v} + H(Mv(0) \tilde{h}' + \rho h_0 W').$$

By (3), $\rho M - M \tilde{h}' \tilde{v} = \rho M - M(\rho - h_0 \tilde{u} H) = Mh_0 \tilde{u} H$, and collecting the factor H gives the claim. $\square$

Lemma 3 (Value). $\rho(f \circ g) = H^2 D$ for an explicit $D \in \mathbb{Z}[X]$.

Proof. Put $B := h_0 HW$, so that $g = MX + B$. By Taylor to second order (Lemma 4) there is $k \in \mathbb{Z}[X]$ with

$$f \circ g = \tilde{f} + \tilde{f}' B + B^2 k.$$

Since $f = hf_1$ we have $\tilde{f} = \tilde{h} \tilde{f}_1 = h_0 H \tilde{f}_1$, and $B^2 = h_0^2 H^2 W^2$, so

$$f \circ g = h_0 H (\tilde{f}_1 + \tilde{f}' W) + H^2 (h_0^2 W^2 k). \quad (4)$$

It remains to extract one more factor of H from the first bracket. Dilating the product rule $f' = h' f_1 + h f_1'$ and multiplying by $\tilde{v}$,

$$\tilde{f}' \tilde{v} = \tilde{h}' \tilde{v} \tilde{f}_1 + \tilde{h} \tilde{f}_1' \tilde{v} = (\rho - h_0 \tilde{u} H) \tilde{f}_1 + h_0 H \tilde{f}_1' \tilde{v},$$

using (3) again. Hence $\rho\widetilde{f}_1 - \widetilde{f}'\widetilde{v} = h_0H(\widetilde{u}\widetilde{f}_1 - \widetilde{v}\widetilde{f}'_1)$, and so

$$\rho(\widetilde{f}_1 + \widetilde{f}'W) = \rho\widetilde{f}_1 + \widetilde{f}'(v(0)H - \widetilde{v}) = HC, \quad C := h_0(\widetilde{u}\widetilde{f}_1 - \widetilde{v}\widetilde{f}'_1) + v(0)\widetilde{f}'.$$

Multiplying (4) by ρ and substituting gives $\rho(f \circ g) = H^2D$ with $D := h_0C + \rho h_0^2W^2k$. $\square$

Lemma 4 (Taylor to second order). *Let R be a commutative ring, $f \in \mathbb{Z}[X]$ and $a, b \in R$. Then there is $k \in R$ with $f(a + b) = f(a) + f'(a)b + b^2k$.*

Proof. Both sides are additive in f and the statement is stable under multiplication by a constant, so it suffices to check $f = X^m$, where the binomial theorem gives $(a + b)^m = a^m + ma^{m-1}b + b^2 \sum_{i \geq 2} \binom{m}{i} a^{m-i} b^{i-2}$. Applied with $R = \mathbb{Z}[X]$, $a = MX$ and $b = B$, this is the form used in Lemma 3. $\square$

4 Descent: removing ρ

Lemmas 2 and 3 give the desired divisibilities only after multiplication by ρ . Removing it is where the normalization $H(0) = 1$ earns its place.

Lemma 5 (Descent). *Let $P, K \in \mathbb{Z}[X]$ and $\rho \in \mathbb{Z}$, $\rho \neq 0$, and suppose $K(0)$ is a unit of $\mathbb{Z}$. If $K \mid \rho P$ in $\mathbb{Z}[X]$ then $K \mid P$ in $\mathbb{Z}[X]$.*

Proof. By strong induction on the number of prime factors of $|\rho|$. If $\rho = \pm 1$ there is nothing to prove. Otherwise choose a prime $p \mid \rho$ and write $\rho = p\rho_1$, and let $\rho P = KD$. Reducing modulo p gives $0 = \overline{K}\overline{D}$ in $\mathbb{F}_p[X]$. Now $K(0) = \pm 1 \not\equiv 0 \pmod{p}$, so $\overline{K} \neq 0$; and $\mathbb{F}_p[X]$ is an integral domain, so $\overline{D} = 0$, that is $D = pD_1$. Cancelling p in the domain $\mathbb{Z}[X]$ leaves $\rho_1 P = KD_1$, and $|\rho_1|$ has one prime factor fewer. $\square$

Corollary 3. $H \mid g'$ and $H^2 \mid f \circ g$, both in $\mathbb{Z}[X]$.

Proof. Apply Lemma 5 to Lemma 2 with $K = H$, whose constant coefficient is 1; and to Lemma 3 with $K = H^2$, whose constant coefficient is likewise 1. $\square$

Remark 2. This is why the construction normalizes $H(0) = 1$ instead of making H monic. Two other routes reach Corollary 3. One: $H(0) = 1$ makes H a unit in the power series ring $\mathbb{Z}[[X]]$, so $(f \circ g)H^{-2}$ lies in $\mathbb{Z}[[X]]$; it also equals $D/\rho \in \mathbb{Q}[X]$; and $\mathbb{Q}[X] \cap \mathbb{Z}[[X]] = \mathbb{Z}[X]$. Two: H is primitive and irreducible, hence prime in $\mathbb{Z}[X]$, and $H \nmid \rho$ by degree. The second route needs §5.2 first, while the induction above and the power series argument need nothing.

5 Nondegeneracy and irreducibility

5.1 Nondegeneracy

Lemma 6. $W \neq 0$, $\deg W \geq 1$, $\deg g = n + \deg W \geq 2$, and $g' \neq 0$.

Proof. Suppose $W = 0$. Then $\widetilde{v} = v(0)H$, and comparing with $h_0H = \widetilde{h}$ gives $h_0\widetilde{v} = v(0)\widetilde{h}$; since dilation is injective, $h_0v = v(0)h$. Then $h \mid h_0v$ in $\mathbb{Q}[X]$, so $v = (v(0)/h_0)h$, and the Bézout relation collapses to $\rho = h(u + (v(0)/h_0)h')$, forcing $\deg h = 0$ and contradicting $n \geq 1$. So $W \neq 0$, and $W(0) = 0$ then gives $\deg W \geq 1$.

Since $M \neq 0$ we have $\deg \widetilde{h} = n$, so $\deg(\widetilde{h}W) = n + \deg W \geq n + 1 \geq 2 > 1 = \deg(MX)$. There is therefore no cancellation in $g = MX + \widetilde{h}W$ and $\deg g = n + \deg W$. Finally $\deg g \geq 2 \geq 1$ and characteristic zero give $g' \neq 0$. $\square$

If the Bézout data is chosen with $\deg v < n$, which is automatic for the subresultant construction, then $\deg W \leq n$ and hence

$$n + 1 \leq \deg g \leq 2n.$$

The lower bound is not an artifact: $H \mid g'$ with $\deg H = n$ forces $\deg g' \geq n$ for *any* solution whatsoever, so no construction can do better. Example 2 attains it.

5.2 Irreducibility

Lemma 7. *H is irreducible in $\mathbb{Z}[X]$.*

Proof. h is primitive and irreducible in $\mathbb{Z}[X]$, hence irreducible in $\mathbb{Q}[X]$ by Gauss's lemma. For $M \neq 0$ the dilation $X \mapsto MX$ is a $\mathbb{Q}$ -algebra automorphism of $\mathbb{Q}[X]$, so $\tilde{h}$ is irreducible in $\mathbb{Q}[X]$; and $H = \tilde{h}/h_0$ differs from it by a unit of $\mathbb{Q}$, so H is irreducible in $\mathbb{Q}[X]$. By Lemma 1, $H(0) = 1$, so H is primitive. Primitive and irreducible over $\mathbb{Q}$ gives irreducible in $\mathbb{Z}[X]$, by Gauss's lemma again. $\square$

This is the only place Gauss's lemma is used, and $\mathbb{Z}$ enters the argument only here and in the prime factorization of Lemma 5. Everything else goes through over any characteristic zero unique factorization domain.

6 The degenerate case

Suppose $X \mid f$. Then

$$g = X^2, \quad H = X$$

settles Theorem 1: $g' = 2X$ is divisible by H ; writing $f = Xf_2$ gives $f \circ g = X^2(f_2 \circ X^2)$, which is divisible by $H^2 = X^2$; and X is irreducible in $\mathbb{Z}[X]$ of degree 1, with $\deg g = 2$.

Note that the case split is on $X \mid f$, not on f being a monomial: this branch covers $f = X(X^2 + 1)$ as well as $f = cX^k$. Equivalently, the split is on whether f has *some* irreducible factor with nonzero constant term.

Proof of Theorem 1. If $X \mid f$, use §6. Otherwise take h , and Bézout data, as in §2, and define M, H, W, g by the table; these lie in $\mathbb{Z}[X]$ by Lemma 1. Then H is irreducible by Lemma 7 with $\deg H = n \geq 1$, the two divisibilities hold by Corollary 3, and $\deg g \geq 2$ by Lemma 6. $\square$

7 Critical values

Only now do roots appear.

Proof of Corollary 2. Let $\alpha \in \overline{\mathbb{Q}}$. If $\alpha = 0$, take $g = X^2$ and $\beta = 0$. Otherwise let h be the primitive integral minimal polynomial of α , so $h(0) \neq 0$, and run §2 with $f = h$, hence $f_1 = 1$. Set

$$\beta := \alpha/M.$$

Then $H(\beta) = h(M\beta)/h_0 = h(\alpha)/h_0 = 0$. Evaluating the Bézout relation at α and using $h(\alpha) = 0$ gives $v(\alpha)h'(\alpha) = \rho$, so

$$W(\beta) = \frac{v(0)H(\beta) - v(\alpha)}{\rho} = -\frac{v(\alpha)}{\rho} = -\frac{1}{h'(\alpha)},$$

which is defined because h is separable. Therefore

$$g(\beta) = M\beta + h(\alpha)W(\beta) = \alpha, \quad g'(\beta) = M + Mh'(\alpha)W(\beta) + h(\alpha)W'(\beta) = M(1 - 1) = 0.$$

So β is a critical point of g with critical value α . The converse was observed at the start of §1. $\square$

Two features of this are worth naming. First, β is the *same* rescaling α/M for every root α of h simultaneously: a single g has all n conjugates as critical values, at the n points α_i/M . Second, the ring at work is $\mathbb{Z}[\beta] = \mathbb{Z}[\alpha/M] \supseteq \mathbb{Z}[\alpha][1/\rho]$. Dividing α by M is what makes the higher Taylor coefficients integral, while $H(0) = 1$ makes β a *unit* in $\mathbb{Z}[\beta]$, and that is what permits trading the constant term of $1/h'(\alpha)$ for higher degree terms.

What does *not* appear is as much to the point: no class group of $\mathbb{Q}(\alpha)$, no unit group $\mathcal{O}_K^\times$, no monogenicity of $\mathcal{O}_K$, and no Thue equation or index form. $\mathbb{Z}[\beta]$ is not an order in $\mathcal{O}_K$ — β is not an algebraic integer — and the units the argument consumes are S -units manufactured by the construction itself.

For a reducible f the statement reduces to the irreducible case for a reason worth stating explicitly, since it is easy to mis-handle: a root α of f is a root of *some* irreducible factor h of f , and the construction run on that h produces a g having α as a critical value. Different factors give different g . It is not the case that one g serves all of a reducible f .

8 Sharpness and scope

H cannot be required monic. The monic form of Theorem 1 is false. Suppose H were monic irreducible with $g(\beta) = \sqrt{2}$ and $g'(\beta) = 0$; then $H^2 \mid g^2 - 2$ in $\mathbb{Z}[X]$. Modulo 2, $\overline{H}^2 \mid \overline{g}^2$ gives $\overline{H} \mid \overline{g}$, so $g = HA + 2B$ for some $A, B \in \mathbb{Z}[X]$. Then

$$g^2 - 2 = H^2 A^2 + 4HAB + 4B^2 - 2,$$

so $H^2 \mid 4HAB + 4B^2 - 2 = 2C$ with $C := 2HAB + 2B^2 - 1$ and $\overline{C} = 1$. As H is monic, division gives $2C = H^2 D$; reducing mod 2 forces $\overline{D} = 0$, so $D = 2E$ and $C = H^2 E$, whence $1 = \overline{H}^2 \overline{E}$ — impossible for $\deg \overline{H} \geq 1$. So $\sqrt{2}$ is not a critical value at any algebraic integer critical point.

Yet $-4\sqrt{2}$ is, and monically: with $g = X^3 - 6X$ and $H = X^2 - 2$,

$$g' = 3(X^2 - 2) = 3H, \quad g^2 - 32 = (X^2 - 2)^2(X^2 - 8) = H^2(X^2 - 8),$$

so $H^2 \mid f \circ g$ for $f = X^2 - 32$. The invariant separating the two cases is the class of $d\alpha$ in $\Omega_{\mathbb{Z}[\beta]/\mathbb{Z}} \cong \mathbb{Z}[\beta]/(H'(\beta))$; it is neither integrality nor a unit condition.

Higher multiplicity is free. Since $H'(\beta)$ is a unit in $\mathbb{Z}[\beta]$, the map $\mathbb{Z}[X] \rightarrow \mathbb{Z}[\beta][Y]/(Y^m)$ sending $X \mapsto \beta + Y$ is surjective, so $H^m \mid f \circ g$ is achievable for every $m \geq 1$. Theorem 1 is the case $m = 2$, which is the one the critical value statement needs.

Hypotheses that are not needed. Nothing requires f to be separable, squarefree or primitive, and $\deg f$ is unbounded; a repeated factor of f only helps. Characteristic zero, on the other hand, is essential, and enters through $h' \neq 0$. In characteristic p with α inseparable the approach dies at $\rho = 0$, and the statement itself degenerates: $g = X^p$ gives $f \circ g = f^p$, but $g' = 0$ has no irreducible factorization.

9 Examples

Each line below is a closed identity, checkable by expansion.

Example 1 ($f = qX - p, p \neq 0$). With $u = 0, v = 1, \rho = q, h_0 = -p, M = -pq, H = q^2X + 1$ and $W = qX$,

$$g = -pq^3X^2 - 2pqX, \quad g' = -2pq(q^2X + 1), \quad f \circ g = -p(q^2X + 1)^2.$$

Example 2 ($f = X^2 + 1$). With $u = 2, v = -X, \rho = 2, h_0 = 1, M = 2, H = 4X^2 + 1$ and $W = X$,

$$g = 4X^3 + 3X, \quad g' = 3(4X^2 + 1), \quad f \circ g = (4X^2 + 1)^2(X^2 + 1).$$

Here $\deg g = 3 = n + 1$ attains the lower bound of §5.1. Note also $\rho = 2$ while $\text{Res}(h, h') = 4$: the reduced resultant is the better choice. The unit group $\mathcal{O}_K^\times = \{\pm 1, \pm i\}$ has rank 0, and it is $\mathbb{Z}[i/2] = \mathbb{Z}[i][1/2]$ that supplies the needed unit $1 + i$.

Example 3 ($f = X^3 - 2$). With $u = 3, v = -X, \rho = -6, h_0 = -2, M = 12, H = 1 - 864X^3$ and $W = -2X$,

$$g = -3456X^4 + 16X, \quad g' = 16(1 - 864X^3).$$

Example 4 ($f = X^3 - X - 1$). The construction gives $M = 23$ and a sextic. A hand-tuned quintic also witnesses the *statement*, and is a useful independent check of it:

$$g = -24334X^5 + 39675X^4 - 26450X^3 + 9085X^2 - 1610X + 118, \quad H = 23X^3 - 23X^2 + 8X - 1,$$

with $g' = -230(23X - 7)H$ and $H^2 \mid f \circ g$, the cofactor having degree 9.

Example 5 (Dedekind's cubic). $f = X^3 - X^2 - 2X - 8$ is the standard example of a field whose ring of integers is not monogenic: 2 is a common index divisor [Ded78]. The construction runs unchanged, which is the point of including it — no step of §2–§4 is sensitive to monogenicity.

It is also where formalization corrected the author. An earlier draft of this material asserted $M = 4024$ here. That is unattainable. Since $M = \rho h_0$ and $h_0 = -8$, it would need $\rho = -503$, the discriminant of the field; but every Bézout constant for this h is even. Writing $h = X^2(X - 1) - 2(X + 4)$ and $h' = 3X^2 - 2(X + 1)$ puts any relation $uh + vh' = c$ into the shape

$$X^2(u(X - 1) + 3v) = c + 2(u(X + 4) + v(X + 1)),$$

and reading off the constant coefficient gives $0 = c + 2(u(X + 4) + v(X + 1))(0)$, so $2 \mid c$. In fact $(h, h') \cap \mathbb{Z} = (1006)$, with certificate

$$(125 - 21X)h + (3 - 44X + 7X^2)h' = -1006,$$

so the smallest attainable $|M|$ is 8048. The relevant invariant is the reduced resultant, not $\text{disc}(K) = -503$; here $\text{disc}(h) = -2012$ and the index is 2.

10 Formalization

The development above has been formalized in Lean 4 [MU21] against Mathlib [The20], in about 1,200 lines. The main statement is

```
1 theorem main (f : Z[X]) (hf : 1 <= f.natDegree) :
2   exists g H : Z[X], Irreducible H /\ H \mid derivative g /\ H^2 \mid f.comp g
```

with a strengthened form carrying $2 \leq \deg g$ and $1 \leq \deg H$, and with Corollaries 1 and 2 proved from it. Every result depends only on the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.

The organization follows the sections above: the dilation and its chain rule (1) first, then `Setup`, a structure carrying the data of §2 with (2) as hypotheses, then Lemmas 2 and 3 as `linear_combination` calls from those hypotheses, then descent, irreducibility and nondegeneracy. Treating H and W as data rather than as quotients, as in Remark 1, is what keeps §3 free of any reasoning about divisibility in $\mathbb{Z}[X]$; the integrality of Lemma 1 is discharged once, in the existence proof for `Setup`.

Three things were genuinely easier on paper than in Lean, and are recorded in case they are useful to someone attempting a similar development. Dilation as a ring homomorphism, and its interaction with `derivative`, is used everywhere and deserves to be isolated first. Lemma 4 is stated in Mathlib for `eval` rather than for `comp`, and has to be transported. And Mathlib has the translation $X \mapsto X + c$ as an algebra equivalence but not the dilation $X \mapsto cX$, which had to be built to prove Lemma 7.

The one place where the formalization changed the mathematics is Example 5, whose erratum is itself machine-checked.

11 Computation

The construction is effective, and a program implementing it accompanies this note. It takes $f \in \mathbb{Z}[X]$, factors it, and for each irreducible factor h returns M, H, W, g and a verification of every hypothesis and conclusion of §2–§5.2 on the actual output.

Two remarks on what the computation costs. The Bézout data is obtained by the extended Euclidean algorithm in $\mathbb{Q}[X]$ followed by content reduction, which yields the generator of $(h, h') \cap \mathbb{Z}$ — the reduced resultant, and so the smallest $|M|$ available. Factoring f is the expensive step and is delegated to FLINT [Har10], which implements van Hoeij’s knapsack recombination [Hoe02] in the improved form of [HNH11]. Root finding, used only for display, is the Aberth–Ehrlich iteration [Abe73], the method underlying MPSolve [BF00].

The coefficients grow quickly, and it is worth being blunt about the size. Since $\deg g = n + \deg W \leq 2n$ and the coefficients of H carry $\rho^j h_0^{j-1}$, the output is large even for small input: $f = x^5 - x - 1$ gives a g of degree 10 with 34-digit coefficients, and $f = x^{10} + 20x^4 + x^3 - 10x^2 + 1$ gives degree 20 with 376-digit coefficients. This is not a defect of the implementation; it is the size of the answer.

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